PART 1: BASELINE FOUNDATION

Standard MDP: Physical Time is Completely Ignored

In standard MDPs, **steps are decision epochs, not seconds on a clock**. A move made after 2 seconds or 30 seconds updates the board state identically.

The Markov Property

The next board distribution depends strictly on current state *s* and action *a*. Earlier computation time carries zero consequence.

$$Pr(s', r \mid s, a, history) = Pr(s', r \mid s, a)$$

Game outcome is evaluated purely by checkmate or draw, with no urgency or time penalty.

Pure Step Count (t = 0, 1, 2)



"The time steps need not refer to fixed intervals of real time; they can refer to arbitrary successive stages of decision making." (Sutton & Barto §3.1)

Key Takeaway: Standard MDP treats every decision as instantaneous with zero elapsed-time consequence.

PART 2: EXAMPLE 1 — BLITZ CHESS

Why SMDP is Superior in Blitz Chess (Avoiding State Explosion)

⚠ In Blitz, elapsed time determines timeout victory or defeat. **Adding clocks into the state explodes the state space**, while **SMDP models duration naturally**.

Naive MDP + Clocks in State

```
s = (board, clock_us, clock_opp)
```

- **Combinatorial State Explosion:** Every single board position is duplicated across hundreds of discrete clock values (180s, 179s, 178s...).
- **Zero Generalization:** The agent cannot transfer tactical learning from a 30s board state to a 20s identical board state.
- **Massive Sample Inefficiency:** Value function convergence slows down exponentially across redundant time-tagged states.

The SMDP Solution: Clean Board + Duration

$$Pr(s', \tau \mid s, a) = Pr(s' \mid s, a) \cdot Pr(\tau \mid s, a, s')$$

- **Compact State Representation:** State remains the pure tactical board: `s = board`.
- **Explicit Time Output (τ):** Move duration τ is predicted directly by the transition dynamics, updating the time budget externally.
- **Direct Time Penalty in Reward:** Elapsed time directly enters the objective function:

$$R(s, a) = E[\ game_reward - \lambda \cdot \tau \mid s, a\ ]$$

The agent learns urgency and time allocation without blowing up the state space!

Bridge to Next: Blitz shows that SMDP prevents clock explosion. Now let's see an even bigger win: **multi-step actions in HRL**!

PART 3: EXAMPLE 2 — CONTINUOUS CONTROL & HRL

Autonomous Driving: Planning Over High-Level Skills with SMDP

Just like Blitz chess, low-level driving control suffers from explosion: **an 8s overtake takes 800 control steps (at 100 Hz)**. SMDP bundles these into one skill.

Low-Level 800-Step MDP Fails

- **Horrendous Planning Horizon:** Millions of pedal commands make trip planning impossible.
- **Exploration Nightmare:** Random exploration over 800 consecutive actions never completes a smooth maneuver.
- **Credit Assignment Lag:** Propagating terminal rewards back 800 steps takes millions of simulation samples.

SMDP Over Skills: 1 Decision = Whole Skill

$$Pr(s', \tau \mid s, a) = Pr(s' \mid s, a) \cdot Pr(\tau \mid s, a, s')$$

- **s':** Destination lane | **τ:** Execution steps (300 vs 800) | **R(s,a):** Accumulated reward

1. Skill (Option)
Packaging 800 pedal commands into 1 high-level decision (e.g. *Overtake*).

2. Skill Reward R(s,a)
Sum of micro-rewards (gas, safety) collected: $r_1 + \gamma r_2 + \ldots + \gamma^{\tau-1} r_\tau$.

3. Discounting (γ^τ)
Penalizes delay: finishing in 3s (γ^{300}) preserves far more value than 8s (γ^{800}).

Total Skill Evaluation: $R(s, a) + \gamma^\tau \cdot Value(s')$

Jumps directly from start to finish while γ^τ guarantees that faster skills achieve higher value!

Key Takeaway: SMDP allows high-level planning over reusable skills while keeping exact accounting of time and reward.

Bradtke & Duff (1994) & Sutton, Precup, Singh (1999)

PART 4: SYNTHESIS & UNIFIED INSIGHT

Unified View: Why SMDP Wins in Both Domains

The Core Insight: Whenever time matters, adding a timer to the state causes **state explosion & sample inefficiency**. SMDP solves both turn-based and continuous problems with temporal abstraction.

Standard MDP

Classical Chess (No Clock)

- **Model:** Pr(s' | s, a)
- **Time Role:** Physical time has zero cost or consequence.
- **Verdict:** Perfect when decisions are purely turn-based and unconstrained by speed.

SMDP in Blitz Chess

Example 1: Time-Budget Allocation

- **Model:** Pr(s', τ | s, a)
- **Why not MDP + Clock?** Avoids duplicating every tactical board across thousands of clock states.
- **SMDP Win:** Keeps board compact while penalizing move duration directly in reward.

SMDP in Autonomous Driving

Example 2: Temporal Abstraction (HRL)

- **Model:** Pr(s', τ | s, a)
- **Why not 800-step MDP?** Solves credit assignment and exploration failure across micro pedal ticks.
- **SMDP Win:** Plans over whole skills in 1 decision while γ^{τ} rewards faster execution.

Summary: SMDP provides the mathematical language for **temporally extended actions**—saving sample efficiency across both games and robotics!

Bradtke & Duff (1994) • Sutton, Precup & Singh (1999)